

# Numerical Integration — One-Page Summary

## The Problem

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Approximate  $\int_a^b f(x) dx$  using a finite number of function evaluations.

## Newton–Cotes (equally spaced nodes, polynomial interpolation)

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Partition  $[a, b]$  into  $N$  panels of width  $h = (b - a)/N$ . On each panel, interpolate  $f$  with a low-degree polynomial and integrate it exactly.

| Rule             | Formula (one panel)                       | Local err.         | Global err.        | Notes                                  |
|------------------|-------------------------------------------|--------------------|--------------------|----------------------------------------|
| Left/Right rect. | $h f(a_i)$ or $h f(b_i)$                  | $\mathcal{O}(h^2)$ | $\mathcal{O}(h)$   | Order 1; crudest                       |
| Midpoint         | $h f(\frac{a_i+b_i}{2})$                  | $\mathcal{O}(h^3)$ | $\mathcal{O}(h^2)$ | Symmetry kills the $h^2$ term          |
| Trapezoidal      | $\frac{h}{2} [f(a_i) + f(b_i)]$           | $\mathcal{O}(h^3)$ | $\mathcal{O}(h^2)$ | Uses endpoints; same order as midpoint |
| Simpson          | $\frac{h}{6} [f(a_i) + 4f(m_i) + f(b_i)]$ | $\mathcal{O}(h^5)$ | $\mathcal{O}(h^4)$ | Parabolic fit; exact for cubics        |

**Key pattern:** local error is  $\mathcal{O}(h^{p+1})$ , global error is  $\mathcal{O}(h^p)$  (one power less, because  $N = (b - a)/h$  errors accumulate).

## Improving Accuracy Without More Nodes

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**Step doubling + Richardson extrapolation.** Compute  $I_1$  (step  $h$ ) and  $I_2$  (step  $h/2$ ) with an  $\mathcal{O}(h^p)$  method. Then:

$$I_{\text{improved}} = I_2 + \frac{I_2 - I_1}{2^p - 1}, \quad \varepsilon \approx \frac{|I_2 - I_1|}{2^p - 1}.$$

Cancels the leading error term; gives both a better estimate *and* a free error estimate.

**Romberg integration.** Apply Richardson extrapolation repeatedly to trapezoidal results at  $h, h/2, h/4, \dots$ . Each level gains +2 orders:  $\mathcal{O}(h^2) \rightarrow \mathcal{O}(h^4) \rightarrow \mathcal{O}(h^6) \rightarrow \dots$

## Adaptive Integration (idea)

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Compare a coarse and fine estimate on each panel (e.g. one Simpson vs. two half-Simpsons). If the difference exceeds a tolerance, subdivide; otherwise accept. Concentrates work where  $f$  varies rapidly.

## Gaussian Quadrature (optimal node placement)

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$$\int_a^b f(x) dx \approx \sum_{k=1}^N w_k f(x_k).$$

Choose  $N$  nodes *and*  $N$  weights ( $2N$  free parameters) so the rule is exact for all polynomials of degree  $\leq 2N - 1$ . Nodes are zeros of orthogonal polynomials (not equally spaced).

**Gauss–Legendre** ( $\rho = 1$  on  $[-1, 1]$ ): nodes = zeros of  $P_N$ ; weights  $w_k = 2/[(1 - x_k^2)(P'_N(x_k))^2]$ .

**Mapping**  $[a, b] \rightarrow [-1, 1]$ :

$$\int_a^b f(x) dx = \frac{b-a}{2} \sum_{k=1}^N w_k f\left(\frac{b-a}{2} t_k + \frac{a+b}{2}\right).$$

**Other families:** Gauss–Chebyshev, Gauss–Hermite, Gauss–Laguerre — same idea, different weight  $\rho(x)$  and domain.

## When to Use What

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| Method                | Best for                                      | Watch out for                                    |
|-----------------------|-----------------------------------------------|--------------------------------------------------|
| Trapezoidal / Simpson | Data on a fixed grid; quick estimates         | Slow convergence for non-smooth $f$              |
| Romberg               | Smooth $f$ , no grid constraint               | Needs $2^j$ -point refinement sequence           |
| Adaptive quadrature   | General-purpose; unknown structure of $f$     | Implementation complexity                        |
| Gauss–Legendre        | Smooth $f$ , free to choose evaluation points | Non-uniform nodes; must recompute if $N$ changes |

**Rule of thumb:** For smooth functions, Gaussian quadrature converges fastest. For data on a grid, Newton–Cotes + Richardson is the practical choice. For unknown or badly behaved  $f$ , use adaptive methods.